EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. If the coefficients of x^7 & x^8 in the expansion of $\left[2+\frac{x}{3}\right]^n$ are equal, then the value of n is :

(A) 15 (B) 45 (C) 55 (D) 56

MBT090

2. If the second term of the expansion $\left[a^{1/13}+\frac{a}{\sqrt{a^{-1}}}\right]^n$ is $14a^{\frac{5}{2}}$, then the value of $\frac{{}^nC_3}{{}^nC_2}$ is

(A) 4 (B) 3 (C) 12 (D) 6

MBT091

3. The value of m, for which the coefficients of the $(2m+1)^{th}$ and $(4m+5)^{th}$ terms in the expansion of $(1+x)^{10}$ are equal, is

(A) 3 (B) 1 (C) 5 (D) 8

MBT092

4. If $k \in R^+$ and the middle term of $\left(\frac{k}{2}+2\right)^8$ is 1120, then value of k is

(A) 3 (B) 2 (C) 1 (D) 4

MBT093

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5. The largest real value for x such that $\sum_{k=0}^{4}\left(\frac{5^{4-k}}{(4-k)!}\right)\left(\frac{x^k}{k!}\right)=\frac{8}{3}$ is -

(A) $2\sqrt{2}-5$ (B) $2\sqrt{2}+5$ (C) $-2\sqrt{2}-5$ (D) $-2\sqrt{2}+5$

MBT094

6. Number of rational terms in the expansion of $\left(\sqrt{2}+\sqrt[4]{3}\right)^{100}$ is :

(A) 25 (B) 26 (C) 27 (D) 28

MBT095

7. In the expansion of $\left(7^{\frac{1}{3}}+11^{\frac{1}{9}}\right)^{6561}$, the number of terms free from radicals is

(A) 730 (B) 729 (C) 725 (D) 750

MBT096

8. If $n \in N$ & n is even, then $\frac{1}{1!(n-1)!}+\frac{1}{3!(n-3)!}+\frac{1}{5!(n-5)!}+\ldots+\frac{1}{(n-1)!1!}=$

(A) 2^n (B) $\frac{2^{n-1}}{n!}$ (C) $2^n n!$ (D) none of these

MBT097

9. Set of value of r for which, ${}^{18}C_{r-2}+2\cdot{}^{18}C_{r-1}+{}^{18}C_r \geq {}^{20}C_{13}$ contains :

(A) 4 element (B) 5 elements (C) 7 elements (D) 10 elements

MBT098

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10. Let $\binom{n}{k}$ represents the combination of 'n' things taken 'k' at a time, then the value of the sum $\binom{99}{97}+\binom{98}{96}+\binom{97}{95}+\ldots+\binom{3}{1}+\binom{2}{0}$ equals -

(A) $\binom{99}{97}$ (B) $\binom{100}{98}$ (C) $\binom{99}{98}$ (D) $\binom{100}{97}$

MBT099

11. The sum of the co-efficient of all the even powers of x in the expansion of $(2x^2-3x+1)^{11}$ is -

(A) 2×6^{10} (B) 3×6^{10} (C) 6^{11} (D) none

MBT100

12. The co-efficient of x^4 in the expansion of $(1-x+2x^2)^{12}$ is

(A) ${}^{12}C_3$ (B) ${}^{13}C_3$ (C) ${}^{14}C_4$ (D) ${}^{12}C_3+3\,{}^{13}C_3+{}^{14}C_4$

MBT101

13. If $\sum_{r=1}^{10} r(r-1)\ {}^{10}C_r = k\times 2^9$, then k is equal to -

(A) 10 (B) 45 (C) 90 (D) 100

MBT102

14. The sum $\frac{\binom{11}{0}}{1}+\frac{\binom{11}{1}}{2}+\frac{\binom{11}{2}}{3}+\ldots+\frac{\binom{11}{11}}{12}$ equals $\left(\text{where } \binom{n}{r} \text{denotes } {}^nC_r\right)$

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(A) $\frac{2^{11}}{12}$ (B) $\frac{2^{12}}{12}$ (C) $\frac{2^{11}-1}{12}$ (D) $\frac{2^{12}-1}{12}$

MBT103

15. Let ${}^0C_0=1$, then $\sum_{m=0}^{n}\sum_{p=0}^{m} {}^nC_m\,.\,{}^mC_p$ is equal to

(A) 2^n-1 (B) 3^n (C) 3^n-1 (D) 2^n

MBT104

16. The last three digits in 10! are :

(A) 800 (B) 700 (C) 500 (D) 600

MBT105

17. Find numerically greatest term in the expansion of $(2+3x)^9$, when $x=3/2$

(A) ${}^9C_6\times 2^9\times\left(\frac{3}{2}\right)^{12}$ (B) ${}^9C_3\times 2^9\times\left(\frac{3}{2}\right)^{6}$ (C) ${}^9C_5\times 2^9\times\left(\frac{3}{2}\right)^{10}$ (D) ${}^9C_4\times 2^9\times\left(\frac{3}{2}\right)^{8}$

MBT106

18. The greatest integer less than or equal to $\left(\sqrt{2}+1\right)^6$ is

(A) 196 (B) 197 (C) 198 (D) 199

MBT107

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If it is known that the third term of the binomial expansion $\left(x+x^{\log_{10} x}\right)^5$ is 10^6 then x is equal to-

(A) 10 (B) $10^{-\frac{5}{2}}$ (C) 100 (D) 5

MBT108

2. In the expansion of $\left(x^3+3\cdot 2^{-\log_{\sqrt{2}}\sqrt{x^3}}\right)^{11}$

(A) there appears a term with the power x^2
(B) there does not appear a term with the power x^2
(C) there appears a term with the power x^{-3}
(D) the ratio of the co-efficient of x^3 to that of x^{-3} is $\frac{1}{3}$

MBT109

3. Let $\left(1+x^2\right)^2(1+x)^n = A_0+A_1x+A_2x^2+\ldots$. If A_0, A_1, A_2 are in A.P. then the value of n is -

(A) 2 (B) 3 (C) 5 (D) 7

MBT110

4. In the expansion $\left(\sqrt[3]{4}+\frac{1}{\sqrt[4]{6}}\right)^{20}$

(A) The number of irrational terms is 19 (B) Middle term is irrational
(C) The number of rational terms is 2 (D) 9th term is rational

MBT111

5. 7^9+9^7 is divisible by :

(A) 16 (B) 24 (C) 64 (D) 72

MBT112

6. Which of the following statement(s) is/are correct ?

(A) $1+\frac{2}{2}+\frac{3}{2^2}+\frac{4}{2^3}+\ldots+\infty=4$

(B) Integral part of $\left(9+4\sqrt{5}\right)^n$, $n \in N$ is even.

(C) $({}^nC_0+{}^nC_1+{}^nC_2+\ldots+{}^nC_n)^2 = 1+{}^{2n}C_1+{}^{2n}C_2+\ldots+{}^{2n}C_{2n}$

(D) $\frac{1}{(3+2x)^2}$ can be expanded as infinite series in ascending powers of x only if $|x|<\frac{2}{3}$.

MBT113

7. ${}^nC_0-2\times 3\,{}^nC_1+3\times 3^2\,{}^nC_2-4\times 3^3\,{}^nC_3+\ldots.+(-1)^n(n+1)\,{}^nC_n 3^n$ is equal to

(A) $2^n\left(\frac{3n}{2}+1\right)$ if n is even (B) $2^n\left(n+\frac{3}{2}\right)$ if n is even

(C) $-2^n\left(\frac{3n}{2}+1\right)$ if n is odd (D) $2^n\left(n+\frac{3}{2}\right)$ if n is odd

MBT114

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8. The sum of the series $\sum_{r=1}^{n}(-1)^{r-1}\cdot {}^{n}C_{r}(a-r)$ is equal to

(A) 5 if $a=5$ (B) –5 if $a=5$

(C) –5 if $a=-5$ (D) 5 if $a=-5$

MBT115

9. Let $P(n)=\sum_{r=0}^{n}\frac{(-1)^{r}r}{r+1}{}^{n}C_{r}$. Now which of the following holds good ?

(A) $|P_{10}|$ is harmonic mean of $|P_9|$ & $|P_{11}|$ (B) $\sum_{r=5}^{10}P(r)P(r-1)=-\frac{6}{55}$

(C) $|P_{10}|$ is arithmetic mean of $|P_9|$ & $|P_{11}|$ (D) $\sum_{r=5}^{10}P(r)P(r-1)=\frac{6}{55}$

MBT116

10. Let $(1+x)^m = C_0 + C_1x + C_2x^2 + C_3x^3 + \ldots + C_mx^m$, where $C_r = {}^{m}C_r$ and $A = C_1C_3 + C_2C_4 + C_3C_5 + C_4C_6 + \ldots + C_{m-2}C_m$, then -

(A) $A \geq {}^{2m}C_{m-2}$ (B) $A < {}^{2m}C_{m-2}$

(C) $A > C_0^2 + C_1^2 + C_2^2 + \ldots C_m^2$ (D) $A < C_0^2 + C_1^2 + C_2^2 + \ldots C_m^2$

MBT117

11. If $\left(1+x+2x^2\right)^{20}=a_0+a_1x+a_2x^2+\ldots.+a_{40}x^{40}$, then $a_0+a_2+a_4\ldots..+a_{38}$ is equal to

(A) $2^{19}\left(2^{30}+1\right)$ (B) $2^{19}\left(2^{20}-1\right)$

(C) $2^{39}-2^{19}$ (D) $2^{39}+2^{19}$

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MBT118

12. If $\left(1+2x+3x^2\right)^{10}=a_0+a_1x+a_2x^2+\ldots+a_{20}x^{20}$, then

(A) $a_1=20$ (B) $a_2=210$

(C) $a_4=8085$ (D) $a_{20}=2^2.3^7.7$

MBT119

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 13 to 14

If $n \in N$ and if $\left(1+4x+4x^2\right)^{n}=\sum_{r=0}^{2n}a_rx^r$, where a_0, a_1, a_2, ..., a_{2n} are real numbers.

13. The unit place of $2\sum_{r=0}^{n}a_{2r}$ can be

(A) 0 (B) 2 (C) 4 (D) 8

MBT123

14. The unit place of $2\sum_{r=1}^{n}a_{2r-1}$ can be

(A) 0 (B) 2 (C) 4 (D) 8

MBT124

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Paragraph for Question No. 15 to 17

Let us consider the binomial expression $(1+x)^n = \sum_{r=0}^{n} a_r x^r$, where a_1, a_2, a_3 are in A.P.

Consider another binomial expression $A = \left(2^{1/3} + 3^{1/4}\right)^{14n}$. The expansion of A contains some rational terms $T_{b_1}, T_{b_2}, \ldots\ldots T_{b_m}$ $(b_1 < b_2 < b_3 < \ldots\ldots < b_m)$, where n is greater than the even prime number.

15. The value of $a_1 + a_2 + a_3$ is –

MBT125

16. The value of $\sum_{r=0}^{n} a_r$ is –

MBT126

17. The smallest value of b_m can be –

MBT127

MATCHING LIST TYPE QUESTION

18.

	List-I	**List-II**
(A)	If $\sum_{r=0}^{10}(r+2^r)\,{}^{10}C_r = 10.2^a + 3^b$, then $(a+b)$ is equal to	(P) 7
(B)	If 4th term in the expansion of $\left(x+(ax)^{-1}\right)^m$, $m \in N$ is $\frac{5}{2}$, then ma is equal to	(Q) 12
(C)	Let λ be the value of numerically greatest term in the expansion of $(1+5x)^{10}$ at $x = 1$, then $\frac{\lambda}{{}^{10}C_9 .5^7}$ is equal to	(R) 19
(D)	If ${}^{n}P_r = 5040\left({}^{n-1}C_r + {}^{n-1}C_{r-1}\right)$, then r is equal to	(S) 25

(A) (A) → (R), (B) → (Q), (C) → (S), (D) → (P)
(B) (A) → (R), (B) → (Q), (C) → (P), (D) → (S)
(C) (A) → (Q), (B) → (R), (C) → (S), (D) → (P)
(D) (A) → (R), (B) → (S), (C) → (Q), (D) → (P)

MBT128

MATRIX MATCH TYPE QUESTION

19.

	Column-I	**Column-II**
(A)	The coefficient of x^{-9} in the expansion of $\left(\frac{x^2}{2}+\frac{2}{x}\right)^9$ is	(P) 1512
(B)	The term independent of x in the expansion of $\left(2x+\frac{3}{x}\right)^6$ is	(Q) 476
(C)	The coefficient of x^{10} in the expansion of $\left(1+x^2-x^3\right)^8$ is	(R) 512
(D)	The coefficient of x^4 in the expansion of $(3x+1)^7\left(1-2x+3x^2\right)$ is	(S) 4320

MBT129

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EXERCISE (S)

1. Let a and b be the coefficient of x^3 in $(1+x+2x^2+3x^3)^4$ and $(1+x+2x^2+3x^3+4x^4)^4$ respectively, then value of $(a-b)$ is

MBT130

2. Let $a=\left(4^{1/401}-1\right)$ and let $b_n={}^nC_1+{}^nC_2\cdot a+{}^nC_3\cdot a^2+\ldots+{}^nC_n\cdot a^{n-1}$, then the value of $\frac{\left(b_{2006}-b_{2005}\right)}{512}$ is

MBT131

3. The coefficients of three consecutive terms of $(1+x)^{n+5}$ are in the ratio $5:10:14$. Then $n=$

MBT134

4. The index n of the binomial $\left(\frac{x}{5}+\frac{2}{5}\right)^n$ if the 9th term of the expansion has uniquely numerically greatest coefficient $(n\in N)$, is

MBT135

5. The sum of the roots (real or complex) of the equation $x^{2001}+\left(\frac{1}{2}-x\right)^{2001}=0$, is

MBT136

6. The coefficient of x^{49} in the polynomial $\left(x-\frac{C_1}{C_0}\right)\left(x-2^2\cdot\frac{C_2}{C_1}\right)\left(x-3^2\cdot\frac{C_3}{C_2}\right)\ldots\left(x-50^2\cdot\frac{C_{50}}{C_{49}}\right)$, where $C_r={}^{50}C_r$, is

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MBT137

7. The coefficient of x^7 in the expansion of $(1-x-x^2+x^3)^6$ is :-

MBT138

8. Let $f(x)=1-x+x^2-x^3+\ldots+x^{16}-x^{17}=a_0+a_1(1+x)+a_2(1+x)^2+\ldots+a_{17}(1+x)^{17}$, then the value of a_2 is

MBT139

9. Let $P=\left(2+\sqrt{3}\right)^5$ and $f=P-[P]$, where $[P]$ denotes the greatest integer function, then the value of $\left(\frac{f^2}{1-f}\right)-720$ is

MBT132

10. If $\left(7+4\sqrt{3}\right)^n=p+\beta$ where n & p are positive integers and β is a proper fraction then value of $(1-\beta)(p+\beta)-1$ is

MBT133

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EXERCISE - JEE (Main) PYQ

1. If the third term in the binomial expansion of $\left(1+x^{\log_2 x}\right)^5$ equals 2560, then a possible value of x is : **[JEE (Main) 2019]**

(1) $2\sqrt{2}$ (2) $\frac{1}{8}$ (3) $4\sqrt{2}$ (4) $\frac{1}{4}$

MBT042

2. The positive value of λ for which the co-efficient of x^2 in the expression $x^2\left(\sqrt{x}+\frac{\lambda}{x^2}\right)^{10}$ is 720, is: **[JEE (Main) 2019]**

(1) $\sqrt{5}$ (2) 4 (3) $2\sqrt{2}$ (4) 3

MBT043

3. Let $(x+10)^{50}+(x-10)^{50}=a_0+a_1x+a_2x^2+\ldots..+a_{50}\,x^{50}$, for all $x\in R$, then $\frac{a_2}{a_0}$ is equal to: **[JEE (Main) 2019]**

(1) 12.50 (2) 12.00 (3) 12.75 (4) 12.25

MBT046

4. The sum of the series $2.{}^{20}C_0+5.{}^{20}C_1+8.\ {}^{20}C_2+11.\ {}^{20}C_3+\ldots+62.\ {}^{20}C_{20}$ is equal to : **[JEE (Main) 2019]**

(1) 2^{24} (2) 2^{25} (3) 2^{26} (4) 2^{23}

MBT048

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5. If the coefficients of x^2 and x^3 are both zero, in the expansion of the expression $(1+ax+bx^2)\,(1-3x)^{15}$ in powers of x, then the ordered pair (a,b) is equal to : **[JEE (Main) 2019]**

(1) (28, 315) (2) (–54, 315) (3) (–21, 714) (4) (24, 861)

MBT050

6. If a, b and c are the greatest value of ${}^{19}C_p$, ${}^{20}C_q$ and ${}^{21}C_r$ respectively, then **[JEE (Main) 2020]**

(1) $\frac{a}{11}=\frac{b}{22}=\frac{c}{21}$ (2) $\frac{a}{10}=\frac{b}{11}=\frac{c}{21}$

(3) $\frac{a}{10}=\frac{b}{11}=\frac{c}{42}$ (4) $\frac{a}{11}=\frac{b}{22}=\frac{c}{42}$

MBT053

7. If α and β be the coefficients of x^4 and x^2 respectively in the expansion of $\left(x+\sqrt{x^2-1}\right)^6+\left(x-\sqrt{x^2-1}\right)^6$, then **[JEE (Main) 2020]**

(1) $\alpha+\beta=60$ (2) $\alpha+\beta=-30$ (3) $\alpha-\beta=-132$ (4) $\alpha-\beta=60$

MBT054

8. The value of $-{}^{15}C_1+2.{}^{15}C_2-3.{}^{15}C_3+\ldots\ldots-15.{}^{15}C_{15}+{}^{14}C_1+{}^{14}C_3+{}^{14}C_5+\ldots.+{}^{14}C_{11}$ is : **[JEE (Main) 2020]**

(1) $2^{16}-1$ (2) $2^{13}-14$ (3) 2^{14} (4) $2^{13}-13$

MBT055

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9. If $n \geq 2$ is a positive integer, then the sum of the series ${}^{n+1}C_2 + 2({}^2C_2 + {}^3C_2 + {}^4C_2 + \ldots + {}^nC_2)$ is: **[JEE (Main) 2021]**

(1) $\frac{n(n-1)(2n+1)}{6}$ (2) $\frac{n(n+1)(2n+1)}{6}$ (3) $\frac{n(2n+1)(3n+1)}{6}$ (4) $\frac{n(n+1)^2(n+2)}{12}$

MBT056

10. The number of elements in the set $\{n \in \{1, 2, 3, \ldots, 100\} \mid (11)^n > (10)^n + (9)^n\}$ is. **[JEE (Main) 2021]**

MBT140

11. Let $[x]$ denote greatest integer less than or equal to x. If for $n \in \mathbb{N}, (1-x+x^3)^n = \sum_{j=0}^{3n} a_j x^j$, then $\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j} + 4\sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j+1}$ is equal to : **[JEE (Main) 2021]**

(1) 2 (2) 2^{n-1} (3) 1 (4) n

MBT142

12. Let $\left(1 + x + 2x^2\right)^{20} = a_0 + a_1x + a_2x^2 + \ldots + a_{40}x^{40}$. then $a_1 + a_3 + a_5 + \ldots + a_{37}$ is equal to **[JEE (Main) 2021]**

(1) $2^{20}(2^{20} - 21)$ (2) $2^{19}(2^{20} - 21)$ (3) $2^{19}(2^{20} + 21)$ (4) $2^{20}(2^{20} + 21)$

MBT143

13. The coefficient of x^{101} in the expression $(5+x)^{500} + x(5+x)^{499} + x^2(5+x)^{498} + \ldots.x^{500}$, $x > 0$, is **[JEE (Main) 2022]**

(1) ${}^{501}C_{101}(5)^{399}$ (2) ${}^{501}C_{101}(5)^{400}$ (3) ${}^{501}C_{100}(5)^{400}$ (4) ${}^{500}C_{101}(5)^{399}$

MBT144

14. Let $n \geq 5$ be an integer. If $9^n - 8n - 1 = 64\alpha$ and $6^n - 5n - 1 = 25\beta$, then $\alpha - \beta$ is equal to: **[JEE (Main) 2022]**

(1) $1 + {}^nC_2(8-5) + {}^nC_3(8^2-5^2) + \ldots + {}^nC_n(8^{n-1} - 5^{n-1})$

(2) $1 + {}^nC_3(8-5) + {}^nC_4(8^2-5^2) + \ldots + {}^nC_n(8^{n-2} - 5^{n-2})$

(3) ${}^nC_3(8-5) + {}^nC_4(8^2-5^2) + \ldots + {}^nC_n(8^{n-2} - 5^{n-2})$

(4) ${}^nC_4(8-5) + {}^nC_5(8^2-5^2) + \ldots + {}^nC_n(8^{n-3} - 5^{n-3})$

MBT146

15. The remainder when $(2021)^{2023}$ is divided by 7 is : **[JEE (Main) 2022]**

(1) 1 (2) 2 (3) 5 (4) 6

MBT147

16. The remainder when $(2021)^{2022} + (2022)^{2021}$ is divided by 7 is **[JEE (Main) 2022]**

(1) 0 (2) 1 (3) 2 (4) 6

MBT148

17. If $\left({}^{40}C_0\right) + \left({}^{41}C_1\right) + \left({}^{42}C_2\right) + \ldots + \left({}^{60}C_{20}\right) = \frac{m}{n}\,{}^{60}C_{20}$, m and n are coprime, then $m + n$ is equal to. **[JEE (Main) 2022]**

MBT149

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18. If $\sum_{k=1}^{31}\left({}^{31}C_k\right)\left({}^{31}C_{k-1}\right)-\sum_{k=1}^{30}\left({}^{30}C_k\right)\left({}^{30}C_{k-1}\right)=\frac{\alpha(60!)}{(30!)(31!)}$, Where $\alpha\in R$, then the value of 16α is equal to

[JEE (Main) 2022]

(1) 1411 (2) 1320 (3) 1615 (4) 1855

MBT150

19. Let the coefficients of x^{-1} and x^{-3} in the expansion of $\left(2x^{\frac{1}{5}}-\frac{1}{x^{\frac{1}{5}}}\right)^{15}$, $x > 0$, be m and n respectively. If r is a positive integer such $mn^2 = {}^{15}C_r \cdot 2^r$, then the value of r is equal to__. [JEE (Main) 2022]

MBT151

20. The constant term in the expansion of $\left(2x+\frac{1}{x^7}+3x^2\right)^5$ is ___. [JEE (Main) 2023]

MBT153

21. If coefficient of x^{15} in the expansion of $\left(ax^3+\frac{1}{bx^{\frac{1}{3}}}\right)^{15}$ is equal to coefficient of x^{-15} in the expansion of $\left(ax^{\frac{1}{3}}-\frac{1}{bx^3}\right)^{15}$, where a and b are positive real numbers, then for each such ordered pair (a, b) :

[JEE (Main) 2023]

(1) $a = b$ (2) $ab = 1$ (3) $a = 3b$ (4) $ab = 3$

MBT154

22. If the coefficient of x^7 in $\left(ax-\frac{1}{bx^2}\right)^{13}$ is equal to coefficient of x^{-5} in expansion of $\left(ax+\frac{1}{bx^2}\right)^{13}$ then a^4b^4 is equal to : [JEE (Main) 2023]

(1) 44 (2) 22 (3) 11 (4) 33

MBT155

23. If the constant term in the binomial expansion of $\left(\frac{x^{\frac{5}{2}}}{2}-\frac{4}{x^{\ell}}\right)^9$ is – 84 and the Coefficient of $x^{-3\ell}$ is $2^{\alpha}\beta$, where $\beta<0$ is an odd number, Then $|\alpha\ell-\beta|$ is equal to _____ [JEE (Main) 2023]

MBT156

24. Let the sixth term in the binomial expansion of $\left(\sqrt{2^{\log_2(10-3^x)}}+\sqrt[5]{2^{(x-2)\log_2 3}}\right)^m$, in the increasing powers of $2^{(x-2)\log_2 3}$, be 21. If the binomial coefficients of the second, third and fourth terms in the expansion are respectively the first, third and fifth terms of an A.P., then the sum of the squares of all possible values of x is. [JEE (Main) 2023]

MBT157

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25. Let the number $(22)^{2022} + (2022)^{22}$ leave the remainder α when divided by 3 and β when divided by 7. Then $(\alpha^2 + \beta^2)$ is equal to **[JEE (Main) 2023]**

(1) 10 (2) 5 (3) 20 (4) 13

MBT158

26. Let $a = 1 + \frac{{}^2C_2}{3!} + \frac{{}^3C_2}{4!} + \frac{{}^4C_2}{5!} + \ldots,$

$b = 1 + \frac{{}^1C_0 + {}^1C_1}{1!} + \frac{{}^2C_0 + {}^2C_1 + {}^2C_2}{2!} + \frac{{}^3C_0 + {}^3C_1 + {}^3C_2 + {}^3C_3}{3!} + \ldots$

Then $\frac{2b}{a^2}$ is equal to _____ **[JEE (Main) 2024]**

MBT162

27. Let $\alpha = \sum_{r=0}^{n} (4r^2 + 2r + 1)\,{}^nC_r$ and $\beta = \left(\sum_{r=0}^{n} \frac{{}^nC_r}{r+1}\right) + \frac{1}{n+1}$. If $140 < \frac{2\alpha}{\beta} < 281$, then the value of n is.

[JEE (Main) 2024]

MBT163

28. If the Coefficient of x^{30} in the expansion of $\left(1 + \frac{1}{x}\right)^6 (1 + x^2)^7 (1 - x^3)^8$; $x \neq 0$ is α, then $|\alpha|$ equals.

[JEE (Main) 2024]

MBT164

29. Remainder when $64^{32^{32}}$ is divided by 9 is equal to ___. **[JEE (Main) 2024]**

MBT165

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30. Let $\alpha = \sum_{k=0}^{n} \left(\frac{\left({}^nC_k\right)^2}{k+1}\right)$ and $\beta = \sum_{k=0}^{n-1} \left(\frac{{}^nC_k\,{}^nC_{k+1}}{k+2}\right)$. If $5\alpha = 6\beta$, then n equals. **[JEE (Main) 2024]**

MBT166

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EXERCISE - JEE (Advanced) PYQ

1. For r = 0, 1,....,10, let A_r, B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}$, $(1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r(B_{10}B_r - C_{10}A_r)$ is equal to - **[JEE (Advanced) 2010]**

(A) $B_{10} - C_{10}$ (B) $A_{10}\left(B_{10}^2 - C_{10}A_{10}\right)$ (C) 0 (D) $C_{10} - B_{10}$

MBT159

2. The coefficients of three consecutive terms of $(1+x)^{n+5}$ are in the ratio 5 : 10 : 14. Then n = **[JEE (Advanced) 2013]**

MBT160

3. Coefficient of x^{11} in the expansion of $(1+x^2)^4(1+x^3)^7(1+x^4)^{12}$ is – **[JEE (Advanced) 2014]**

(A) 1051 (B) 1106 (C) 1113 (D) 1120

MBT161

4. Let m be the smallest positive integer such that the coefficient of x^2 in the expansion of $(1+x)^2 + (1+x)^3 + \ldots\ldots\ldots + (1+x)^{49} + (1+mx)^{50}$ is $(3n+1)\,{}^{51}C_3$ for some positive integer n. Then the value of n is **[JEE (Advanced) 2016]**

MBT075

5. Let $X = ({}^{10}C_1)^2 + 2({}^{10}C_2)^2 + 3({}^{10}C_3)^2 + \ldots + 10({}^{10}C_{10})^2$, where ${}^{10}C_r, r \in \{1,2,\ldots,10\}$ denote binomial coefficients. Then, the value of $\frac{1}{1430}X$ is ____. **[JEE (Advanced) 2018]**

MBT076

6. Suppose $det\begin{bmatrix} \sum_{k=0}^{n} k & \sum_{k=0}^{n} {}^{n}C_k k^2 \\ \sum_{k=0}^{n} {}^{n}C_k k & \sum_{k=0}^{n} {}^{n}C_k 3^k \end{bmatrix} = 0$, holds for some positive integer n.

Then $\sum_{k=0}^{n} \frac{{}^{n}C_k}{k+1}$ equals. **[JEE (Advanced) 2019]**

MBT077

7. For non-negative integers s and r, let $\binom{s}{r} = \begin{cases} \frac{s!}{r!(s-r)!} & \text{if } r \le s, \\ 0 & \text{if } r > s. \end{cases}$

For positive integers m and n, let $g(m,n) = \sum_{p=0}^{m+n} \frac{f(m,n,p)}{\binom{n+p}{p}}$

Where for any non-negative integer p, $f(m,n,p) = \sum_{i=0}^{p} \binom{m}{i}\binom{n+i}{p}\binom{p+n}{p-i}$

Then which of the following statements is/are TRUE? **[JEE (Advanced) 2020]**

(A) $g(m,n) = g(n,m)$ for all positive integers m, n

(B) $g(m,n+1) = g(m+1,n)$ for all positive integers m, n

(C) $g(2m,2n) = 2g(m,n)$ for all positive integers m, n

(D) $g(2m,2n) = (g(m,n))^2$ for all positive integers m, n

MBT078

8. Let a and b be two nonzero real numbers. If the coefficient of x^5 in the expansion of $\left(ax^2 + \frac{70}{27bx}\right)^4$ is equal to the coefficient of x^{-5} is equal to the coefficient of $\left(ax - \frac{1}{bx^2}\right)^7$, then the value of $2b$ is

[JEE (Advanced) 2023]

MBT079

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	B	B	A	B	A	B	C	D
Que.	11	12	13	14	15	16	17	18		
Ans.	B	D	B	D	B	A	A	B		

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	B,C,D	A,B	A,B,C,D	A,C	A,C	A,C	A,C	A,D	B,D
Que.	11	12	13	14	15	16	17	18		
Ans.	B,C	A,B,C	A,B	A,D	63.00	128.00	93.00	A		
Que.	19									
Ans.	(A → R; B → S; C → Q; D → P)									

EXERCISE (S)

1. 0 **2.** 2 **3.** 6 **4.** 12.00 **5.** 500.00
6. –22100.00 **7.** –144.00 **8.** 816.00 **9.** 2 **10.** 0

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	2	4	2	1	4	3	2	2	96
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	1	3	3	1	102	1	5	1080
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	2	98	4	2	8	5	678	1	10

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8
Ans.	D	6	C	5	646	6.20	A,B,D	3

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